a_compact_commercial_air-source_heat_pump

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	**a_compact_commercial_airsource_heat_pump**
Industry Domain	**commercial_hvac**
Engine Version	**PDF Engine v2.1 (Full Detail)**
Target Quantity	**Not specified**

ARCHITECTURE AT A GLANCE

MODULES	BILL OF MATERIALS ROWS	IDENTIFIED RISKS
9	**41**	**27**

ECONOMICS

EST. UNIT COST	TARGET CEILING	HEADROOM
£8,152.50	**None**	**N/A**

TOTAL NON-RECURRING ENGINEERING	UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING
£5,400.00	**£13,552.50**

Contents

1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Providing reliable, high-efficiency space heating and domestic hot water for retrofitted or newly built small commercial properties.

Mission Statement

To accelerate the decarbonization of small commercial buildings by providing an affordable, highly efficient, and future-proof R290 split air-source heat pump.

Target Customers

Small commercial building owners, HVAC contractors, retail spaces, and small office operators in the UK.

Why Now (Market Timing)

The UK government's aggressive push to phase out fossil fuel boilers, rising energy costs, and stringent F-Gas regulations make high-efficiency, very-low-GWP heating solutions immediately necessary.

Full Synthesis Report

The UK commercial heating market is undergoing a rapid transition driven by changing regulations, the impending phase-down of high-GWP HFC refrigerants under F-Gas regulations, and government incentives aimed at decarbonizing the built environment. Developing a 30kW commercial air-source heat pump utilizing R290 (propane) is a highly strategic move given its near-zero Global Warming Potential (GWP of 3) and excellent thermodynamic properties, which can comfortably support a COP of >= 3.5 at A2/W35. However, the exact architectural requirement for a split system (indoor and outdoor unit) rather than a monobloc introduces critical safety engineering challenges. Because R290 is a highly flammable A3 class refrigerant, routing refrigerant lines into an indoor unit requires stringent adherence to BS EN 378 safety limits, necessitating advanced leak detection, fail-safe ventilation, and potential ATEX-rated internal components to prevent explosive accumulation in occupied spaces. Technically, the specification of a stainless steel heat exchanger ensures longevity and corrosion resistance, while an R290-optimized scroll compressor and an efficient axial fan will minimize acoustic footprints. The primary business challenge lies in the unit economics: achieving a manufactured unit cost of £4,500 at a low volume of 500 units per year. Overcoming this requires leveraging economies of scale via standardized

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

off-the-shelf compressors and heat exchangers. Furthermore, MCS (Microgeneration Certification Scheme) certification is mandatory for the UK market to allow customers to access commercial grants, making early engagement with testing bodies for BS EN 14511 compliance the top priority.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.2 Market & Competitors

Competitor Landscape

Vaillant

Product	**aroTHERM plus (Cascaded)**
Pricing Strategy	**£8,000+ (for dual unit cascade approach)**
Technical Specs	**15kW units (cascaded to 30kW), Monobloc, R290, COP ~4.9 at A7/W35.**

STRENGTHS

Exceptional brand trust in UK, extremely quiet operation, well-established MCS track record.

WEAKNESSES

Monobloc only; requires more external footprint and a complex cascade controller for 30kW output.

DIFFERENTIATION ANGLE

Providing a single 30kW split unit to minimize footprint and simplify installation logic compared to dual-unit cascades.

Mitsubishi Electric

Product	**Ecodan CAHV Series**
Pricing Strategy	**£9,500+**
Technical Specs	**Large capacity commercial air source, scroll twin compressors, typically higher GWP refrigerants transitioning to R32/R290.**

STRENGTHS

Market leader in commercial HVAC, highly advanced control ecosystem, exceptional reliability.

WEAKNESSES

High capital cost, older models are bulky, slow transition to pure R290 in smaller commercial brackets.

DIFFERENTIATION ANGLE

Aggressive price point (£4,500 target cost allowing competitive retail pricing) and native R290 design aimed specifically at the lower end of the commercial spectrum.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Clivet

Product	**Thunder Series (Commercial Heat Pump)**
Pricing Strategy	**£7,000**
Technical Specs	**R290, 30kW+, Scroll compressors, High-efficiency axial fans.**

STRENGTHS

Deep expertise in commercial capacities, robust industrial-grade builds, early adopters of R290.

WEAKNESSES

Lower brand recognition among generic UK plumbers and small MEP contractors; sparse local service network.

DIFFERENTIATION ANGLE

Hyper-focused on UK MCS compliance and specific features designed for tight UK retrofits, rather than generic European commercial markets.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	**£0.00**
Max Mass	**-**
Target Process	
Target Material	
Tolerance Target	
Production Quantity	

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
UK Microgeneration Certification Scheme (MCS) Product Standards	Regulatory Standard	2026	Outlines the exact testing and certification criteria required to sell eligible heat pumps in the UK.
BSRIA UK Heat Pump Market Report	Market Research	2026	Provides competitive landscape data, volume expectations, and pricing models for commercial HVAC.
UK F-Gas Regulations & Phasedown Schedule	Government Policy	2026	Validates the market shift towards R290 by detailing the ban timeline for high-GWP refrigerants like R410A.
Copeland Scroll Compressor Application Guidelines for R290	Technical Datasheet	2026	Essential engineering data for integrating a scroll compressor at 30kW capacity with propane.
CIBSE Guide B: Heating, Ventilating, Air Conditioning and Refrigeration	Engineering Guide	2026	Provides the load calculations and system design requirements for UK small commercial buildings.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Brief — Quality Score

Section Evaluation: Brief

Composite Score: 5/10

Factual Integrity — are numbers accurate and claims grounded in reality?	7/10	Claims regarding the UK government's push for decarbonization and F-Gas regulations driving R290 adoption are factually accurate and grounded in current market reality.
Visual Clarity — is the layout clean, readable, with proper hierarchy?	6/10	The key-value pair layout is simple and readable, but leaving multiple fields entirely blank visually detracts from the professional quality and hierarchy.
Linguistic Precision — is the language professional, specific, free of filler?	7/10	The completed fields utilize professional, industry-specific terminology (e.g., 'very-low-GWP', 'decarbonization') with minimal filler.
Logical Coherence — does this section flow from the previous one?	5/10	While the first half flows logically from mission to market context, the abrupt shift to completely empty engineering parameters breaks the flow and reading experience.
Actionability — can the founder take concrete next steps from this section?	3/10	Founders cannot make concrete engineering, manufacturing, or financial decisions from this section because all actionable metrics (cost, mass, materials) are missing.
Requirement Completeness	2/10	More than half of the required fields (Process, Material, Tolerance, Quantity, Compliance) are entirely empty, rendering the brief incomplete.
Constraint Capture	2/10	Fails to capture key engineering constraints. Both Cost Ceiling and Max Mass are 'not set', and no manufacturing tolerances or compliance standards are defined.
Unambiguity	5/10	The product vision is clear and unambiguous, but the physical requirements are completely vague and ambiguous due to the lack of specified thresholds or materials.

Recommended Changes for Next Round:

• Implement a validation hook in the pipeline that halts generation or throws a warning if mandatory technical fields (Process, Material, Tolerance, Compliance) evaluate to null or empty strings.

• Add an LLM fallback prompt to automatically estimate or suggest generic placeholder constraints (e.g., relevant ISO/EN standards for 'Compliance' based on the product type) when user input is missing.

• Update the templating engine logic to handle missing variables gracefully, replacing blank outputs with 'TBD - Pending Input' or conditionally hiding fields that are not applicable to the current phase.

• Add a required 'Technical Specifications' function that generates or prompts for core performance metrics (e.g., COP target, output kW range, max refrigerant charge) to populate the empty fields.

• Implement a constraint-generation subroutine that, based on the mission and use case, automatically suggests and populates initial values for Cost Ceiling, Max Mass, and Compliance standards (e.g., IEC 60335-2-40, UK Building Regs) for user validation.

Research — Quality Score

Section Evaluation: Research

Composite Score: 6/10

Factual Integrity — are numbers accurate and claims grounded in reality?	6/10	Deterministic assessment
Visual Clarity — is the layout clean, readable, with proper hierarchy?	6/10	Deterministic assessment
Linguistic Precision — is the language professional, specific, free of filler?	6/10	Deterministic assessment
Logical Coherence — does this section flow from the previous one?	6/10	Deterministic assessment
Actionability — can the founder take concrete next steps from this section?	6/10	Deterministic assessment
Technical Depth	6/10	Deterministic assessment
Source Recency	6/10	Deterministic assessment
Relevance to Design Decisions	6/10	Deterministic assessment

2.1 MCS 007

Heat Pump Product Certification Standard

Status: not-started | **Owner: Compliance Engineer**

Standard Summary

Specific requirements for the certification of heat pumps in the UK.

Applicability Assessment

> Entire unit and its commercial documentation.

Engineering Design Impact

Dictates energy efficiency labeling rules, weather compensation features, and testing thresholds.

EVIDENCE REQUIRED

Accredited test reports and factory production control audits.

GAP ACTION

Identify and book an MCS-accredited testing laboratory.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.2 BS EN 378

Refrigerating systems and heat pumps - Safety and environmental

Status: not-started | Owner: Mechanical Engineering Lead

Standard Summary

Safety definitions for flammable refrigerants and charge limits.

Applicability Assessment

> Indoor unit and refrigerant piping.

Engineering Design Impact

Crucially impacts indoor unit design due to R290 (A3) flammability; mandates leak sensors and ventilation.

EVIDENCE REQUIRED

Design risk assessment, charge limit calculations, failure mode tests.

GAP ACTION

Calculate maximum R290 system charge limit based on minimum expected room volume.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.3 BS EN 14511

Air conditioners, liquid chilling packages and heat pumps

Status: not-started | Owner: Thermal Systems Engineer

Standard Summary

Standardized test conditions for rating performance (COP) and capacity.

Applicability Assessment

> Thermal performance validation.

Engineering Design Impact

Drives the sizing of the stainless steel heat exchanger, compressor, and axial fan to achieve COP >= 3.5 at A2/W35.

EVIDENCE REQUIRED

Performance data from a certified climatic chamber test.

GAP ACTION

Develop an alpha prototype for internal pre-compliance thermal testing.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.4 UKCA / Electrical Equipment (Safety) Regulations

UK Low Voltage Directive Equivalent

Status: not-started | Owner: Electrical Engineer

Standard Summary

Ensures electrical equipment is safe for use.

Applicability Assessment

> All electrical components, primarily compressor drives, fan motors, and controllers.

Engineering Design Impact

Requires specific grounding, dielectric strength, and ignition-proof components (ATEX) near R290.

EVIDENCE REQUIRED

Technical file with LVD test results.

GAP ACTION

Audit electrical BOM for ignition-proof compliance.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

2.5 ErP Directive (UK Equivalent)

Ecodesign for Energy-Related Products

Status: not-started | **Owner: Acoustics/NVH Engineer**

Standard Summary

Sets minimum energy performance and maximum acoustic emissions.

Applicability Assessment

> Outdoor unit acoustics and seasonal efficiency (SCOP).

Engineering Design Impact

Requires optimized axial fan blade design and acoustic jacketing for the scroll compressor.

EVIDENCE REQUIRED

Sound power level test report and SCOP declaration.

GAP ACTION

Perform CFD and acoustic simulation on the outdoor unit enclosure.

Data Sources:
[LLM] Gemini 3.1 Pro — standards extraction

Regulatory — Quality Score

Section Evaluation: Regulatory

Composite Score: 7/10

Factual Integrity — are numbers accurate and claims grounded in reality?	7/10	Deterministic assessment
Visual Clarity — is the layout clean, readable, with proper hierarchy?	7/10	Deterministic assessment
Linguistic Precision — is the language professional, specific, free of filler?	7/10	Deterministic assessment
Logical Coherence — does this section flow from the previous one?	7/10	Deterministic assessment
Actionability — can the founder take concrete next steps from this section?	7/10	Deterministic assessment
Standard Relevance	7/10	Deterministic assessment
Compliance Gap Analysis	7/10	Deterministic assessment
Certification Path Clarity	7/10	Deterministic assessment

3. Sizing & Spatial Optimisation

INFEASIBLE DESIGN ALLOCATION
The current sizing constraints are not satisfied. Module requirements exceed the available envelope.

System Envelope

KIND	RULES DOMAIN	FEASIBLE
generic_room	commer-cial_hvac	No

Internal Dimensions (W×D×H)	**5000 × 5000 × 3000 mm**
Total Internal Floor Area	**25 m²**
Total Internal Volume	**75 m³**
Available Floor Budget	**21.25 m²**

Module Dimension Allocations

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALE
outdoor_refrigeration_circuit	2236 × 2236 × 2400	5	floor	mass_pro-portional
outdoor_evaporator_coil	2236 × 2236 × 2400	5	floor	mass_pro-portional
outdoor_fan_assembly	2236 × 2236 × 2400	5	floor	mass_pro-portional
outdoor_heat_exchanger	2236 × 2236 × 2400	5	floor	mass_pro-portional
outdoor_power_electronics	2236 × 2236 × 2400	5	floor	mass_pro-portional

Data Sources:
[Deterministic] Sizing solver and envelope constraints

outdoor_chassis_acoustics	2236 × 2236 × 2400	5	floor	mass_proportional
r290_safety_system	2236 × 2236 × 2400	5	floor	mass_proportional
indoor_hydrobox	2236 × 2236 × 2400	5	floor	mass_proportional
indoor_ui_controller	2236 × 2236 × 2400	5	floor	mass_proportional

Identified Conflicts

- Total module footprint (45.0m2) exceeds envelope budget (21.3m2)

Optimisation Recommendations

- Consider a larger envelope
- Optimize module footprint

Data Sources:
[Deterministic] Sizing solver and envelope constraints

Sizing — Quality Score

Section Evaluation: Sizing

Composite Score: 6/10

Factual Integrity — are numbers accurate and claims grounded in reality?	6/10	Deterministic assessment
Visual Clarity — is the layout clean, readable, with proper hierarchy?	6/10	Deterministic assessment
Linguistic Precision — is the language professional, specific, free of filler?	6/10	Deterministic assessment
Logical Coherence — does this section flow from the previous one?	6/10	Deterministic assessment
Actionability — can the founder take concrete next steps from this section?	6/10	Deterministic assessment
Calculation Accuracy	6/10	Deterministic assessment
Margin Analysis	6/10	Deterministic assessment
Assumption Traceability	6/10	Deterministic assessment

4.1 Refrigerant Compression and Metering Assembly

Lead Time: 14 weeks | Status: draft

Module Purpose

Circulates and compresses the R290 (propane) refrigerant to execute the vapor-compression cycle.

Why it matters to the system

> It dictates the raw thermal capacity and compressor efficiency (COP target >= 3.5), operating as the mechanical core of the heat pump.

Detailed Technical Description

This module represents the thermodynamic heart of the heat pump. Utilizing a robust, inverter-driven scroll compressor optimized for R290, it elevates the low-pressure, low-temperature refrigerant vapor from the evaporator into a high-pressure, high-temperature state. This hot gas is then delivered to the brazed plate heat exchanger to reject heat into the hydronic loop. The modular piping sub-assembly also includes the primary Electronic Expansion Valve (EEV) and reversing valve, which are critical for dynamically controlling superheat and managing periodic defrost cycles when operating below freezing ambient conditions.

Designing this module requires careful attention to the A3 flammability classification of propane. All brazed joints must be heavily validated to eliminate micro-leaks, and electrical connections at the compressor terminals must be secured in a way that minimizes ignition risks. Furthermore, ensuring that the compressor is acoustically jacketed and mounted on precise vibration isolators is critical to achieving the stringent ErP acoustic emission targets without compromising structural integrity.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Low-pressure R290 vapor
- 400-V 3-phase AC power (via inverter)

- High-pressure/high-temperature R290 gas
- Modulated refrigerant flow

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.1.2 Specifications & Risk Profile

Expected Key Parts

- R290-Optimized scroll compressor
- Electronic Expansion Valve (EEV)
- 4-way Reversing valve
- Suction line accumulator
- High/Low pressure transducers

Identified Failure Modes

- Liquid slugging causing mechanical wear to the compressor scrolls
- Micro-leaks at compressor brazing joints leading to loss of R290 charge
- EEV stepper motor stalling resulting in incorrect superheat and efficiency collapse
- Reversing valve solenoid sticking during defrost cycle transition

Open Questions

- Exact vibration harmonic frequencies at 30kW peak load
- Optimal accumulator sizing to completely eliminate liquid return under extreme rapid defrosting

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.1.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
outdoor_refrigeration_circuit-001	R290-Optimized Scroll Compressor	Purchased	-	£800.00
outdoor_refrigeration_circuit-002	Electronic Expansion Valve (EEV)	Purchased	-	£150.00
outdoor_refrigeration_circuit-003	4-Way Reversing Valve	Purchased	-	£80.00
outdoor_refrigeration_circuit-004	Suction Line Accumulator	Purchased	-	£40.00
outdoor_refrigeration_circuit-005	High/Low Pressure Transducers	Purchased	-	£60.00
			Module Total Cost:	**£1,130.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.1.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The thermodynamic design is highly efficient and sound, but utilizing R290 (propane) introduces severe safety and manufacturability hurdles. Transitioning this to mass production requires ATEX-certified manufacturing zones, precision automated brazing, and critical design modifications to the acoustic jacket.

Identified Issues

RISK

Acoustic jacketing around the compressor poses a severe explosion hazard as it can trap leaked R290 gas near the electrical terminals.

Suggestion: Redesign the acoustic jacket to be breathable or naturally ventilated so gas cannot pool, while still meeting ErP acoustic targets. Ensure the compressor terminal box is IP65/ATEX certified.

MANUFACTURABILITY

Manual brazing introduces excessive operator variance, resulting in unacceptable rates of micro-leaks for highly flammable R290.

Suggestion: Implement automated induction brazing or robotic flame brazing coupled with a 100% Helium mass spectrometry hard-vacuum leak testing station.

TOLERANCES

Automated brazing requires highly consistent capillary action, which will fail if pipe fit-up tolerances are too loose.

Suggestion: Specify strict +/- 0.05mm tolerances on all copper tube swaging and expansion joints to ensure optimal brazing gaps.

Data Sources:
[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

ENGINEERING

EEV stepper motor stalls and reversing valve sticking are often caused by improper mounting orientations combined with system debris.

Suggestion: Mandate vertical mounting orientations for the EEV and reversing valve in the 3D layout, and ensure a system filter-drier is installed upstream of the EEV.

RISK

Manufacturing facility will require significant capital expenditure to handle A3 refrigerants safely.

Suggestion: Factor in the cost and floor space for an ATEX-compliant Class 1 Div 2 hazardous manufacturing zone for charging and end-of-line testing.

Strategic Recommendations

- Redesign the acoustic jacket to prevent gas pooling (e.g., ventilated bottom, breathable dense foam).
- Replace manual brazing operations with automated induction/robotic brazing in the assembly process.
- Implement 100% Helium leak detection but of a vacuum chamber prior to R290 charging.
- Lock the mounting orientations of the EEV and Reversing Valve to vertical to prevent mechanical binding/stalling.
- Ensure all electrical connections inside the compressor envelope are sealed with intrinsically safe, ignition-proof connectors.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.2 Air-to-Refrigerant Evaporator Module

Lead Time: 10 weeks | Status: draft

Module Purpose

Absorbs sensible and latent thermal energy from ambient air to vaporize liquid R290.

Why it matters to the system

> Maximizing surface area and airflow efficiency here is the primary driver of extracting free heat from the air, directly dictating the heat pump's seasonal COP.

Detailed Technical Description

The evaporator module is a massive heat exchange matrix composed of specialized copper micro-groove tubes mechanically expanded into intricately folded, corrugated aluminum fins. As the system's axial fans pull immense volumes of outdoor air across this coil, the low-pressure liquid R290 evaporating inside the copper tubes absorbs the thermal energy. The aluminum fins are coated with a hydrophilic layer designed to optimize condensate water drainage and resist biological growth or environmental corrosion, which is vital for long-term commercial installations.

A primary engineering challenge at the A2/W35 condition (2°C air) is the rapid accumulation of frost on the evaporator fins, which chokes airflow and drastically reduces the Coefficient of Performance (COP). Fin spacing must be strategically optimized—wide enough to delay frost bridging between fins, but dense enough to achieve the necessary surface area for 30kW of heat extraction within a restricted footprint. The module also includes an electric defrost trace heating cable integrated into the basepan to ensure melted ice drains freely and does not refreeze as a persistent ice block.

SYSTEM INPUTS

- Ambient air
- Low-pressure liquid R290

SYSTEM OUTPUTS

- Chilled ambient air
- Low-pressure R290 vapor

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.2.2 Specifications & Risk Profile

Expected Key Parts

- Copper microtubing
- Corrugated aluminium fins
- Hydrophilic anti-corrosion coating
- Defrost base-heater cable

Identified Failure Modes

- Fin damaging during assembly or transit, impeding local airflow
- Galvanic corrosion between copper tubes and aluminium fins over years of operation
- Frost bridging failure mode where defrost cycles fail to clear dense ice build-up
- Base-pan drain freezing leading to coil encasement in solid ice

Open Questions

- Airflow maldistribution across the massive 30kW fin face
- Exact defrost cycle durations needed for high-humidity UK winters without penalizing SCOP

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.2.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
outdoor_evaporator_coil-001	Copper Micro-Groove Tubing	Purchased	-	£40.00
outdoor_evaporator_coil-002	Corrugated Aluminum Fins	Manufactured	-	£60.00
outdoor_evaporator_coil-003	Hydrophilic Anti-Corrosion Coating	Purchased	-	£40.00
outdoor_evaporator_coil-004	Defrost Base Heater Cable	Purchased	-	£40.00
			Module Total Cost:	**£180.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.2.4 Engineering Review

Review by Fang

WARN

Verdict Summary
The evaporator uses a proven micro-groove Cu/Al architecture that minimizes R290 charge, but requires strict handling mitigations for the large 30kW coil format and stringent leak testing due to R290 flammability.

Identified Issues

RISK

R290 is a highly flammable A3 refrigerant. Galvanic corrosion between dissimilar metals (Cu/Al) or micro-cracking during tube expansion could cause fatal leaks.

Suggestion: Implement 100% inline Helium vacuum leak testing. Use pre-coated aluminum fin stock (epoxy + hydrophilic) and apply a protective conformal/epoxy coating to the copper U-bend hairpins.

ENGINEERING

Basepan heaters only prevent ice build-up in the drain tray. At 2°C (A2/W35), the coil itself will experience severe frost bridging, which the base heater cannot clear.

Suggestion: Explicitly integrate a reverse-cycle (hot gas) defrost circuit routing into the tube matrix, heavily weighted toward the bottom circuitry to ensure complete frost melt.

MANUFACTURABILITY

A 30kW evaporator coil will be exceptionally large and heavy, making the corrugated aluminum fins highly susceptible to crushing during assembly, handling, and transit.

Suggestion: Design heavy-gauge galvanized steel or aluminum end-plates with integrated lifting/hoisting points to isolate load paths away from the fin pack.

Data Sources:
[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

TOLERANCES

Mechanical expansion of micro-groove tubes requires extremely tight tolerance control on the expansion bullet diameter to prevent wall thinning or internal groove deformation.

Suggestion: Specify an expansion ratio tolerance of +/- 0.05mm and utilize automated torque/thrust-monitored expansion equipment on the line.

Strategic Recommendations

- Incorporate heavy-duty rigid end-plates with lifting eyes to prevent fin damage during factory handling.
- Implement isolated hot-gas reverse-cycle defrost loops within the lower quarter of the coil.
- Specify pre-coated aluminum fin stock that combines an anti-corrosion epoxy primer with the hydrophilic top-coat.
- Mandate 100% automated Helium vacuum mass spectrometry leak testing at the end of the module's assembly line due to R290 safety requirements.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.3 Axial Fan & Shroud Module

Lead Time: 8 weeks | Status: draft

Module Purpose

Draws large volumes of ambient air uniformly across the evaporator coil.

Why it matters to the system

> Essential for providing the thermal source (air volume) and highly influential on both ErP efficiency limits and acoustic compliance.

Detailed Technical Description

To achieve 30kW of thermal output, a tremendous volume of ambient air must be pulled through the evaporator coil. This module consists of dual high-efficiency Electronically Commutated (EC) axial fans, their aerodynamically swept fan blades, and a specifically molded shroud (bell-mouth) that maximizes static pressure while minimizing turbulent air shedding. EC motors are selected over traditional AC motors because they offer continuous, precise modulation via PWM signals, adapting perfectly to structural heat loads and contributing significantly to passing the stringest UK ErP directive efficiency requirements.

Beyond raw airflow, this module is the primary source of the outdoor unit's acoustic emissions. Given the strict noise regulations in urban small-commercial environments, the fan blades must feature an owl-wing or serrated trailing edge design to break up tonal noise frequencies. The cowling design and the physical mounting of the fan struts on rubber isolators are hyper-critical to prevent structural resonance from propagating through the Z275 steel outer casing.

SYSTEM INPUTS

SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- 230 V AC or 48 V DC power (for EC motors)
  - PWM control signal

- High-velocity air exhaust
  - Fan tachometer telemetry

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.3.2 Specifications & Risk Profile

Expected Key Parts

- Electronically Commutated (EC) axial fans
- Aerodynamically optimized cowling/shroud
- Vibration isolation motor mounts
- Bird guard/safety grille

Identified Failure Modes

- EC motor bearing seizure due to long-term weather exposure
- Acoustic hum/resonance crossing legal threshold decibel limits
- PWM controller failure inside the motor housing
- Ice accumulation on the fan blades causing extreme out-of-balance vibrations

Open Questions

- Precise sound power level at maximum RPM in a production chassis
- Tolerance and limits of fan structural mounts under UK high-wind conditions

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.3.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
outdoor_fan_assembly-001	Electronically Commutated (EC) Axial Fans	Purchased	-	£250.00
outdoor_fan_assembly-002	Aerodynamically Optimized Cowling/Shroud	Manufactured	-	£60.00
outdoor_fan_assembly-003	Vibration Isolation Motor Mounts	Purchased	-	£40.00
outdoor_fan_assembly-004	Bird Guard/Safety Grille	Manufactured	-	£60.00
			Module Total Cost:	**£410.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.3.4 Engineering Review

Review by Fang

WARN

Verdict Summary
This dual EC fan module correctly addresses ErP efficiency and urban noise constraints, but presents significant manufacturing and reliability risks regarding ice-induced imbalance and tight fan-to-shroud tip clearances.

Identified Issues

RISK

Ice accumulation on the fan blades can cause severe out-of-balance conditions, leading to catastrophic vibration that the rubber isolators cannot absorb.

Suggestion: Implement a software-controlled reversible fan cycle for active ice shedding, and consider using blade materials with slight flexibility to cause passive ice fracturing.

TOLERANCES

To maximize static pressure, the clearance between the swept blades and the bell-mouth shroud must be kept very tight. However, large molded shrouds are susceptible to thermal expansion and warping, risking blade rub.

Suggestion: Add stiffening ribs to the shroud molding and specify a tight profile tolerance on the bell-mouth. Ensure the assembly jig centers the motor perfectly before torquing down the mounts.

MATERIALS

The EC motors, including their internal PWM controllers, are directly exposed to driving rain, snow, and extreme temperature variations.

Suggestion: Specify IP66-rated EC motors with double-sealed bearings and conformal coated PCBs for the PWM controllers to prevent moisture-induced failure.

Data Sources:
[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

MANUFACTURABILITY

The complex 'owl-wing/serrated' blade design requires high-precision injection molding and must be dynamically balanced to prevent tonal hum.

Suggestion: Institute a required 100% End-of-Line (EOL) dynamic balancing and acoustic test station.

Strategic Recommendations

- Integrate a fan reversal sequence during the system defrost cycle to actively blow off accumulated ice and snow.
- Specify a UV-stabilized, glass-filled Polypropylene (PP) or Polyamide (PA) for the shroud and blades to maintain structural rigidity over a wide temperature range.
- Establish a strict concentricity and runout tolerance budgets during DFM, backed by precision assembly fixtures.
- Require IP66-rated motors with potted or conformal-coated electronic modules to eliminate PWM controller failure.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.4 Brazed Plate Heat Exchanger (Condenser)

Lead Time: 12 weeks

Status: draft

Module Purpose

Transfers heat from the high-pressure R290 gas into the circulating hydronic (water/glycol) loop.

Why it matters to the system

> It creates the physical separation boundary between the hazardous outdoor R290 circuit and the safe indoor water circuit, serving as the system's thermal delivery payload mechanism.

Detailed Technical Description

The condenser module utilizes a 316L Stainless Steel Brazed Plate Heat Exchanger (BPHE). In this highly compact, counter-flow component, the high-pressure, superheated R290 refrigerant gas transfers its sensible and latent heat directly into the circulating water/glycol mixture returning from the building's indoor heating loop. The counter-flow design ensures maximum thermal transfer efficiency, dropping the refrigerant into a sub-cooled liquid state while elevating the water temperature up to 75°C—ideal for retrofitting older commercial buildings that rely on traditional high-temperature radiators.

To prevent catastrophic destruction, the module incorporates a strict flow switch and thermal sensors. If the hydronic pump fails or the water circuit loses pressure, the high-temperature refrigerant could rapidly boil stagnant water, causing overpressurization, or alternatively, during defrost actions, it could freeze the water, rupturing the internal steel plates. A thick, hermetically sealed polyurethane insulation jacket is wrapped around the entire BPHE to prevent parasitic heat loss to the cold ambient outdoor air.

SYSTEM INPUTS

SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- High temperature R290 gas
  - Cold return water/glycol from indoor loop

- Subcooled liquid R290
  - Hot supply water (up to 75°C)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.4.2 Specifications & Risk Profile

Expected Key Parts

- 316L Stainless Steel Brazed Plate Heat Exchanger (BPHE)
- Polyurethane thermal insulation jacket
- Water flow switch
- Water supply/return thermistors

Identified Failure Modes

- Internal plate rupture due to water freezing during reverse cycle defrosts
- Fouling/scaling on the water side reducing heat transfer efficiency
- Thermal insulation tearing causing major parasitic energy losses
- Refrigerant leak into the water loop through compromised internal brazing

Open Questions

- Water quality impact of hard water areas on the BPHE over a 15-year lifecycle
- Minimal required freeze-protection flow rate required during a -15C ambient defrost

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.4.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
outdoor_heat_ex-changer-001	316L Stainless Steel Brazed Plate Heat Exchanger (BPHE)	Purchased	-	£600.00
outdoor_heat_ex-changer-002	Polyurethane Thermal Insulation Jacket	Purchased	-	£15.00
outdoor_heat_ex-changer-003	Water Flow Switch	Purchased	-	£40.00
outdoor_heat_ex-changer-004	Water Supply/Return Thermistors	Purchased	-	£40.00
			Module Total Cost:	**£695.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.4.4 Engineering Review

Review by Fang

FAIL

Verdict Summary

The module presents a catastrophic safety risk due to the potential for highly flammable R290 to leak into the indoor hydronic loop in the event of an internal plate rupture. While thermally efficient, the proposal requires immediate design remediation for A3 refrigerant safety compliance.

Identified Issues

RISK

Internal plate rupture (due to freezing or fatigue) in a single-wall BPHE will force high-pressure, highly flammable R290 into the low-pressure hydronic loop, carrying explosive gas into the occupied building.

Suggestion: Mandate a Double-Wall Brazed Plate Heat Exchanger (BPHE) with a vented interstitial space to safely vent any leaked R290 to the outdoor atmosphere.

MANUFACTURABILITY

A hermetically sealed polyurethane jacket applied permanently over the BPHE prevents End-of-Line (EOL) rework and lifetime servicing. If fittings leak during testing, the unit must be scrapped.

Suggestion: Replace poured/sprayed PU with a molded, two-piece interlocking EPP (Expanded Polypropylene) jacket.

ENGINEERING

Rapid freezing during reverse-cycle defrost is a known risk. Relying solely on a flow switch is inadequate due to sensor latency and mechanical failure rates.

Suggestion: Implement a required pump run-on cycle during defrost mode, verified glycol concentration minimums, and consider a localized trace heater.

Data Sources:
[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

MATERIALS

The brazing material for the 316L SS plates is unspecified. Uninhibited glycol or improper water chemistry at 75°C can rapidly corrode standard copper braze joints.

Suggestion: Specify 99.9% Copper or Nickel brazing based on water loop requirements, and strictly define the allowable site water chemistry.

TOLERANCES

Pushing water/glycol mixtures to 75°C substantially accelerates scale precipitation from mineral-heavy water, rapidly reducing the micro-channel tolerances in the BPHE.

Suggestion: Require an upstream Y-strainer and magnetic dirt separator to protect the BPHE channels from fouling.

Strategic Recommendations

- Upgrade the BOM to include a double-wall BPHE to isolate R290 from the building hydronic loop.
- Transition to a 2-piece molded EPP foam insulation enclosure for improved manufacturability, DFM, and serviceability.
- Add a hardware freeze-protection protocol in the control logic (active pump override during defrost) rather than strictly relying on flow switch cutoff.
- Detail brazing alloy specifications and establish strict warranty limits tied to primary loop water chemistry.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.5 Power Electronics & Inverter Drive Enclosure

Lead Time: 16 weeks | Status: draft

Module Purpose

Conditions power, drives the compressor/fans, and hosts the main operational logic within an ATEX-compliant enclosure.

Why it matters to the system

> It controls the efficiency map of the entire system via the inverter and contains the largest potential ignition sources which must be neutralized to allow safe use of R290.

Detailed Technical Description

This sub-assembly acts as the neurological and power distribution hub for the outdoor unit. Operating on a commercial 400V 3-phase grid supply, it houses the main inverter drive that precisely manipulates the frequency and voltage sent to the scroll compressor, allowing the heat pump to smoothly scale its thermal output from 10kW up to 30kW as building demand shifts. It also contains the main microprocessor logic board that processes inputs from dozens of thermistors, pressure transducers, and flow sensors, running the complex PID loops that maintain optimum thermodynamic superheat.

Because R290 (A3 class) is highly explosive, the physical design of this module is dictated heavily by BS EN 60335-2-40 and ATEX safety directives. The entire module must be housed in an inherently non-sparking, hermetically sealed sub-enclosure to ensure that if a refrigerant leak occurs in the outer chassis, no gas can penetrate the electrical bay to be ignited by an arcing contactor. EMI/RFI filtration is also dense here to prevent the massive power cables from broadcasting electrical noise back onto the commercial building's power grid.

SYSTEM INPUTS

SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- 400 V 3-Phase AC Grid Power
- Sensor Telemetry (Temp, Pressure, Flow)

- Variable-frequency 3-phase power to compressor
- PWM to fans
- Actuation signals (EEV, Solenoids)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.5.2 Specifications & Risk Profile

Expected Key Parts

- 3-Phase Inverter Drive PCB
- Main Logic Controller (e.g., Carel)
- ATEX-rated hermetically sealed enclosure
- EMI/RFI electrical filter
- High-voltage contactor relays

Identified Failure Modes

- Inverter IGBT module thermal failure from poor heatsinking
- Electrical arcing igniting trapped R290 gas in the event of enclosure seal failure
- Logic board capacitor degradation over 10+ years causing erratic resets
- EMI interference disrupting local commercial communication networks

Open Questions

- Thermal load inside the sealed electrical box during mid-summer peak cooling (if reverse cycle enables cooling)
- Exact EMI filter sizing required to pass UKCA/CE EMC directives

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.5.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
outdoor_power_electronics-001	3-Phase Inverter Drive PCB	Purchased	-	£400.00
outdoor_power_electronics-002	Main Logic Control Controller	Purchased	-	£180.00
outdoor_power_electronics-003	ATEX-Rated Hermetically Sealed Enclosure	Manufactured	-	£120.00
outdoor_power_electronics-004	EMI/RFI Electrical Filters	Purchased	-	£40.00
outdoor_power_electronics-005	High-Voltage Contactor Relays	Purchased	-	£40.00
			Module Total Cost:	**£780.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.5.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The module presents a critical engineering paradox: dissipating the high thermal loads of a 30kW-capable IGBT inverter while maintaining a strict, potentially hermetic, ATEX-compliant seal against R290 gas.

Identified Issues

ENGINEERING

Enclosing high-power IGBTs and EMI filters in a fully sealed enclosure will cause massive internal heat buildup, directly triggering the identified thermal failure modes.

Suggestion: Implement a thermal passthrough design where the IGBT heatsink extends out of the enclosure into the main fan slipstream, sealed with an ATEX-rated compressible EMI/gas gasket.

MANUFACTURABILITY

True 'hermetic' sealing is incompatible with standard 400V 3-phase power cable routing. Assembly operators will struggle to maintain ATEX-certified gas-tight boundaries when running dozens of I/O and thick power cables.

Suggestion: Require EX-d or EX-e rated multi-core cable glands with highly specific torque tooling on the assembly line, or use pre-potted wire harnesses.

MATERIALS

Capacitor degradation over 10+ years is virtually guaranteed if ambient temperatures inside the sealed enclosure drift above 70C.

Suggestion: Specify AEC-Q200 automotive-grade capacitors rated for 125C minimum, and utilize conformal coating to protect against any trace moisture trapped during sealing.

Data Sources:
[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

RISK

High-voltage contactor relays are inherent sparking devices. If the enclosure seal fails over time due to thermal cycling, R290 ingress will result in a catastrophic ignition.

Suggestion: Consider solid-state relays (SSRs) to eliminate physical arcing entirely, or incorporate an internal micro-ventilation/sniffer system that trips power before the LEL (Lower Explosive Limit) is reached.

Strategic Recommendations

- Re-design the inverter thermal management to utilize an externally protruding heat sink with an ATEX-certified interface gasket.
- Replace mechanical high-voltage contactor relays with non-sparking Solid State Relays (SSRs) to inherently eliminate the ignition source.
- Upgrade all logic board electronic capacitors to 125°C/10,000-hour rated components to survive the elevated internal enclosure temperatures.
- Mandate automated torque drivers and traceability logging for all EX-rated cable glands during enclosure assembly to prevent human error in sealing.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.6 Structural Frame and Acoustic Jacketing

Lead Time: 6 weeks | Status: draft

Module Purpose

Housings all outdoor components, providing structural rigidity, weather protection, and absorbing acoustic emissions.

Why it matters to the system

> It dictates the physical durability of the product, serviceability for the installer, and is the primary tool to meet strict aesthetic and noise regulations.

Detailed Technical Description

The chassis provides the outer shell of the 30kW machine, designed explicitly for ground or flat-roof mounting in British weather conditions. Constructed from heavy-grade Z275 galvanized sheet metal coated in UV-resistant polyester powder paint, it must resist corrosion in damp, coastal, or urban environments. The structural frame dictates the unit's manufacturability, supporting the heavy 250kg weight of the compressor, fans, and heat exchangers while ensuring maintenance access doors allow easy serviceability by commercial HVAC plumbers.

Acoustic suppression is intrinsically woven into this module. The UK ErP directive imposes strict sound power limits. Therefore, the chassis utilizes mass-loaded vinyl blankets wrapped directly around the compressor body, and closed-cell foam lining the interior sheet metal to absorb mid-to-high frequency whine from the inverter and compressor scrolls. The basepan is sloped and designed to channel hundreds of liters of water generated during winter defrost cycles safely out of the unit to avoid creating dangerous ice slicks around the installation area.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Physical load of internal components
- Environmental weather elements

- Dampened acoustic waves
- Direct condensate drainage

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.6.2 Specifications & Risk Profile

Expected Key Parts

- Z275 Galvanized steel structural frame
- Polyester powder-coated exterior panels
- Mass-loaded vinyl acoustic compressor wraps
- Condensate drainage basin
- Rubber isolation feet

Identified Failure Modes

- Galvanic or environmental rust compromising load-bearing walls after 5 years
- Harmonic resonance from the compressor multiplying chassis vibration into a loud rattle
- Base-pan drain holes undersizing, leading to overflowing internal water that freezes
- Acoustic foam degrading over time and shedding into the fan path

Open Questions

- Vibrational transfer function through the frame to the mounting points at various compressor speeds
- Optimal paint thickness and bake time to achieve 15-year salt-spray guarantees at 500 units/year economies

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.6.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
outdoor_chassis_acoustics-001	Z275 Galvanized Steel Structural Frame	Manufactured	-	£60.00
outdoor_chassis_acoustics-002	Polyester Powder-Coated Exterior Panels	Manufactured	-	£60.00
outdoor_chassis_acoustics-003	Mass-Loaded Vinyl Acoustic Compressor Wraps	Purchased	-	£800.00
outdoor_chassis_acoustics-004	Condensate Drainage Basepan	Manufactured	-	£60.00
outdoor_chassis_acoustics-005	Rubber Isolation Feet	Purchased	-	£40.00
			Module Total Cost:	**£1,020.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.6.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The structural frame and acoustic jacketing provide a solid foundation, but the design carries notable risks regarding coastal cut-edge corrosion, basepan freezing, and long-term acoustic foam degradation.

Identified Issues

ENGINEERING

Sloping the basepan and oversizing drain holes is insufficient for UK winter defrost cycles; water will freeze upon contact with the cold steel basepan before draining, building an internal ice block.

Suggestion: Integrate a thermostatically controlled trace heating cable into the basepan to ensure defrost run-off remains liquid until fully evacuated.

MATERIALS

Z275 galvanized steel, when sheared or punched, leaves exposed cut edges that powder coating struggles to cover entirely. In coastal UK environments, this will lead to premature edge rust.

Suggestion: Upgrade plain Z275 to a Zinc-Aluminum-Magnesium coating (e.g., Magnelis ZM310) which offers galvanic self-healing on cut edges, or introduce an epoxy primer dip prior to powder coating.

RISK

Adhesive-backed closed-cell foam is prone to delamination and degradation over a 10-15 year lifecycle due to temperature swings, humidity, and compressor oil vapors.

Suggestion: Specify a ruggedized facing (e.g., Mylar or reinforced aluminum foil) onto the continuous face of the foam, and use mechanical weld-pins alongside high-grade acrylic adhesives.

Data Sources:
[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

MANUFACTURABILITY

At 250kg, manual handling on the factory floor and during field installation is a high risk. Lifting and moving constraints could deform the lower chassis if not rigidly structured.

Suggestion: Incorporate formed forklift pockets directly into the base structural rails and provide designated top-lifting eyelet points to prevent twisting of the chassis during rigging.

ENGINEERING

Relying purely on rubber isolation feet to prevent harmonic ringing across heavy sheet metal spans is optimistic. Compressor vibration will likely excite the large panel surfaces.

Suggestion: Perform FEA modal analysis to identify resonant frequencies and press stiffening ribs or 'X' breaks into large flat sheet metal panels to shift their natural frequencies out of the compressor's operating envelope.

Strategic Recommendations

- Add a basin and trace heater element interlocked with the unit's defrost cycle controls.
- Transition sheet metal spec from Z275 to a self-healing Zinc-Aluminium-Magnesium alloy (Magnelis) for vastly superior coastal corrosion resistance.
- Add mechanically affixed (weld pins/clips) holding points for acoustic foam and specify vapor/oil-proof backing film.
- Apply stamped ribs to the large flat external panels to stiffen the structure against low-frequency harmonic resonance.
- Ensure all exposed sheet metal edges (lips) are hemmed for safety during assembly and to minimize raw edges prone to rust.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.7 R290 Leak Monitoring & Safety Purge System

Lead Time: 10 weeks

Status: draft

Module Purpose

Actively detects high-pressure refrigerant leaks inside the chassis and prevents atmospheric accumulation from reaching the Lower Flammability Limit (LFL).

Why it matters to the system

> Without it, commercial deployment of R290 systems is illegal and catastrophic explosion risks exist.

Detailed Technical Description

Due to the A3 classification (highly flammable) of R290, adherence to BS EN 378 requires failsafe systems to prevent any leaked gas from entering building spaces or reaching combustible concentrations. Since this is a hydronic split, the safety risk is isolated entirely to the outdoor unit. However, should an internal pipe rupture near the non-ATEX fan motors or a service technician's tool, an explosion risk is present. This module utilizes Non-Dispersive Infrared (NDIR) gas sensors mounted locally at the base of the cabinet (since propane is heavier than air and pools at the bottom) to continuously sniff for ppm-level gas signatures.

Upon detection of a leak over a defined threshold (e.g., 20% of the LFL), the intrinsically safe relay board immediately kills power to the compressor, seals the solenoid lock-out valves to isolate the refrigerant charge, and drops power from non-ATEX components. Simultaneously, it engages an ATEX-rated emergency purge fan to forcibly ventilate the pooled propane out into the open atmosphere, dispersing it instantly and signaling an unrecoverable hard-fault to the BMS system.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- Air samples from inside the electrical/compressor bay

- Emergency shutdown signal
- Forced ATEX exhaust fan actuation

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.7.2 Specifications & Risk Profile

Expected Key Parts

- Infrared combustible gas sensor (NDIR)
- Intrinsically safe relay board
- Emergency purge exhaust fan
- Solenoid safety lock-out valves

Identified Failure Modes

- NDIR sensor drift causing false positive shutdowns
- Sensor masking from dirt/dust preventing actual leak detection
- Purge fan failing during a genuine leak event
- Failure of safety relay, leaving main power live during a massive gas dump

Open Questions

- Calibration lifecycle of the NDIR sensors before they require mechanical replacement
- Exact airflow geometry required to purge the cabinet within 30 seconds

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.7.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
r290_safety_system-001	Infrared Combustible Gas Sensors (NDIR)	Purchased	-	£60.00
r290_safety_system-002	Intrinsically Safe Relay Board	Purchased	-	£40.00
r290_safety_system-003	Emergency Purge Exhaust Fan	Purchased	-	£250.00
r290_safety_system-004	Solenoid Safety Lock-Out Valves	Purchased	-	£80.00
			Module Total Cost:	**£430.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.7.4 Engineering Review

Review by Fang

WARN

Verdict Summary

While the active safety module provides robust leak response, enclosing an outdoor R290 unit tightly enough to require forced purging contradicts passive safety best practices and introduces severe manufacturing and assembly complexities regarding intrinsically safe (IS) wiring and End-of-Line testing.

Identified Issues

ENGINEERING

Outdoor R290 units should primarily rely on design-integrated passive natural ventilation (louvers, open bases) to disperse heavy gas leaks, per IEC 60335-2-40. Relying on active purge systems adds high-risk single points of failure.

Suggestion: Redesign the cabinet enclosure for passive bottom ventilation to naturally disperse propane, treating the active purge as redundancy rather than the primary mitigation strategy.

MANUFACTURABILITY

Intrinsically Safe (IS) relays and circuitry require strict electrical separation, specific creepage/clearance distances, and dedicated routing (usually blue wiring), slowing down assembly and requiring specialized QA.

Suggestion: Implement dedicated, segregated raceways and poka-yoke harness connectors for all IS wiring to prevent assembly errors and cross-routing with high-voltage leads.

RISK

Standard NDIR sensors installed in the base of an outdoor unit are highly vulnerable to water pooling, mud, dust, and insects, which can mask the sensor and completely disable leak detection.

Suggestion: Specify IP65+ rated sensors equipped with sintered PTFE hydrophobic filters and built-in heating elements to prevent condensation and environmental masking.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

MANUFACTURABILITY
Verifying the safety circuit's function requires exposing the sensors to a calibrated test gas (e.g., 20% LFL equivalent) during End-of-Line (EOL) production testing, requiring specialty gas handling on the factory floor.

Suggestion: Design an automated, localized 'bump-test' station at the EOL run-test cell with integrated exhaust extraction to safely validate the NDIR sensors and safety relay sequence.

MATERIALS
The solenoid lock-out valves themselves represent a potential ignition source if they are actuated during a leak event. Standard solenoids are unacceptable.

Suggestion: Ensure the solenoid coils are ATEX/Ex rated (e.g., Exm encapsulated or Exia) and are Normally Closed (NC) so they fail-safe to a sealed state upon power loss.

Strategic Recommendations

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

- Reevaluate the sheet metal cabinet architecture to maximize passive, bottom natural ventilation to prevent pooling, reducing total reliance on the ATEX fan.
- Implement segregated, color-coded IS wire routing into the bill of materials and assembly work instructions to guarantee proper separation from mains power.
- Deploy a multi-gas EOL test station on the manufacturing line to functionally bump test the NDIR sensors and ensure the main contactors drop out correctly.
- Upgrade NDIR sensors to severe-duty, environmentally compensated models with sintered metal or PTFE filtering to prevent sensor masking.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.8 Indoor Hydronic Pump Station (Hydrobox)

Lead Time: 6 weeks

Status: draft

Module Purpose

Circulates heating water from the outdoor unit through the commercial building, managing water pressure, air flushing, and safety venting.

Why it matters to the system

> It translates the outdoor unit's thermal output into usable building heat while completely isolating building occupants from R290 flammability risks.

Detailed Technical Description

To maintain strict safety per EN 378, this heat pump utilizes a hydronic split architecture. The entire explosive R290 circuit sits outside, pushing only hot water into the building. The 'indoor unit' is therefore a high-end pump station (hydrobox). Housed within a sleek wall-mounted casing indoors, it manages the fluid dynamics of the heating system. A commercial-grade, variable-speed circulation pump pushes water back to the outdoor heat exchanger to absorb heat, and routes it to the building's domestic hot water cylinders or space heating buffer tanks.

Crucially, this module handles hydronic stability. A large internal expansion vessel absorbs the volumetric expansion of the water as it is heated from 10°C to 75°C, preventing pipe rupture. An Automatic Air Vent (AAV) purges microscopic air bubbles to prevent cavitation inside the pump volute, while a magnetic internal filter captures iron oxide sludge returning from old retrofitted commercial radiators before it can damage the outdoor brazed plate heat exchanger. It represents a highly standardized, easy-to-install plumbing block for generic heating contractors.

SYSTEM INPUTS

SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

• Hot water supply from outdoor unit • Building return water • 230-V AC Power	• Pressurized/Heated water to building emitters (Radiators/UFH)

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.8.2 Specifications & Risk Profile

Expected Key Parts

- Variable speed circulating pump (e.g., Grundfos/Wilo)
- Diaphragm expansion vessel (e.g., 18-24 liters)
- Pressure Relief Valve (PRV, 3-bar)
- Automatic Air Vent (AAV)
- Magnetic system filter

Identified Failure Modes

- Circulation pump cavitation and failure from trapped system air
- Expansion vessel diaphragm tearing, causing the system to over-pressurize and vent all water
- PRV leaking continuously causing massive pressure drops in the building loop
- Magnetic filter choking the flow rate due to heavy sludge accumulation in old buildings

Open Questions

- Vessel sizing adequacy: 30kW systems may require supplementary external vessel volumes depending on building size
- Pump acoustic vibration transfer into the interior wall

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.8.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
indoor_hydrobox-001	Variable Speed Circulating Pump	Purchased	-	£250.00
indoor_hydrobox-002	Diaphragm Expansion Vessel	Purchased	-	£40.00
indoor_hydrobox-003	Pressure Relief Valve (PRV, 3-bar)	Purchased	-	£80.00
indoor_hydrobox-004	Automatic Air Vent (AAV)	Purchased	-	£40.00
indoor_hydrobox-005	Magnetic System Filter	Purchased	-	£40.00
			Module Total Cost:	**£450.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.8.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The hydrobox concept effectively mitigates R290 indoor risk, but specified components—particularly the 18-24L expansion vessel and internal filter—are undersized for the water volumes and sludge loads typically found in commercial retrofits.

Identified Issues

ENGINEERING

An 18-24L expansion vessel is sufficient for residential systems but grossly undersized for commercial applications. A commercial retrofitted loop with old cast-iron radiators will have a massive system volume, guaranteeing PRV weeping/venting if limited to a 24L internal vessel.

Suggestion: Remove the structural burden of the expansion vessel from the wall-mounted hydrobox. Supply it as a separate floor-standing unit, or mandate installation of a site-specific vessel calculated by the heating contractor.

RISK

Old commercial radiators will shed immense amounts of iron oxide sludge. A compact internal magnetic filter will quickly choke, severely restricting flow to the outdoor heat exchanger and causing high-pressure lockouts.

Suggestion: Mandate an external, high-capacity commercial dirt and magnetic separator (e.g., SpiroTrap) on the return line, and add a differential pressure sensor across the internal hydrobox lines to trigger pre-emptive cleaning warnings.

MANUFACTURABILITY

Packaging a commercial-grade pump, 24L vessel, brass PRV, AAV, and filter into a 'sleek wall-mounted casing' creates extreme assembly density. This will slow down factory routing of copper/stainless pipework and make field service (like pump replacement) nearly impossible.

Suggestion: Increase cabinet dimensions by at least 15% to allow for standardized wrench clearances. Use modular slide-out trays for the pump and filter assemblies.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

MATERIALS

The PRV and AAV seals, along with the expansion vessel diaphragm, must be rated for continuous thermal cycling up to 75°C. Standard commercial rubbers may degrade and harden over time at these upper limits.

Suggestion: Ensure all wetted elastomers are high-grade EPDM, specifically rated and tested for 90°C to provide a safety margin.

TOLERANCES

Torqueing heavy 1-inch or 1.25-inch commercial brass fittings inside a sheet-metal chassis risks warping the interior frame if alignment tolerances are not perfectly maintained.

Suggestion: Design the internal pipework sub-assembly on a rigid jig outside the main chassis, and drop it in as a single structural element.

Strategic Recommendations

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

- Recalculate internal expansion vessel requirements, pivot to an external floor-standing vessel to save internal space and cover actual commercial volume needs.
- Implement an electronic differential pressure (DP) switch across the hydronic circuit to detect filter clogging before pump cavitation occurs.
- Require automated end-of-line hydrostatic pressure testing at 4.5 bar (1.5x PRV rating) on the factory floor with helium tracer or visual inspection.
- Redesign the internal layout for Serviceability Level Assessment (SLA) to ensure a technician can extract the pump motor head or clean the filter in under 10 minutes without unmounting the box.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

4.9 Human-Machine Interface & Building Management Control

Lead Time: 14 weeks | Status: draft

Module Purpose

Provides user scheduling, diagnostics, cascaded system orchestration, and integration with commercial Building Management Systems (BMS).

Why it matters to the system

> It determines the ultimate intelligence of the heat pump, ensures MCS compliance (grant funding eligibility), and dictates the ease of interfacing for high-end commercial clients.

Detailed Technical Description

Mounted on or near the indoor hydrobox, the HMI and BMS module serves as the functional brain-trust for the commercial operators. Commercial heating systems demand robust scheduling and diagnostic data. This module runs the 'weather compensation' curves strictly mandated by the Microgeneration Certification Scheme (MCS), dynamically adjusting the targeted flow temperatures based on real-time external weather to maximize SCOP (Seasonal Coefficient of Performance). Furthermore, since 30kW can be limiting for larger commercial sites, this controller is designed with cascade capabilities, allowing up to four 30kW units to talk to each other and load-balance a 120kW thermal demand seamlessly.

Importantly, the UK commercial sector requires integration into legacy or centralized control setups. By housing native Modbus and BACnet protocols, this subsystem easily plugs into existing Building Management Systems. A Wi-Fi/LTE transceiver provides cloud-based telemetry, allowing remote diagnostic contractors to view pressure and temperature graphs off-site, reducing costly truck rolls and facilitating predictive maintenance insights.

SYSTEM INPUTS	SYSTEM OUTPUTS

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

- User touch inputs
- Outdoor temperature sensors
- Modbus/BACnet instructions

- Control targets to outdoor logic board
- Cloud telemetry
- Diagnostic UI screens

Data Sources:

[LLM] Gemini 3.1 Pro — module decomposition

4.9.2 Specifications & Risk Profile

Expected Key Parts

- LCD capacitive touchscreen interface
- Microcontroller processing baseboard
- Modbus/BACnet transceiver chips
- Wi-Fi/Cellular connectivity module
- Weather compensation thermistor inputs

Identified Failure Modes

- Firmware bug causing weather compensation logic to command inappropriate freezing or boiling setpoints
- Wi-Fi/Cellular connectivity failures in basement plant rooms
- LCD/screen failure/delamination in humid mechanical environments
- Modbus communication bus collisions when cascading multiple units

Open Questions

- Complexity limits of the user interface for generic, untrained commercial building managers
- Cellular signal attenuation inside typical concrete commercial plant rooms

Data Sources:
[LLM] Gemini 3.1 Pro — module decomposition

4.9.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
indoor_ui_con-troller-001	LCD Capacitive Touch Screen Interface	Purchased	-	£40.00
indoor_ui_con-troller-002	Microcontroller Processing Baseboard	Purchased	-	£180.00
indoor_ui_con-troller-003	Modbus/BACnet Transceiver Chips	Purchased	-	£40.00
indoor_ui_con-troller-004	Wi-Fi/Cellular Connectivity Module	Purchased	-	£40.00
indoor_ui_con-troller-005	Weather Compensation Thermistor Inputs	Purchased	-	£40.00
			Module Total Cost:	**£340.00**

Data Sources:
[LLM] Gemini 3.1 Pro — part generation
[Deterministic] Cost model calculation

4.9.4 Engineering Review

Review by Fang

WARN

Verdict Summary
The HMI and BMS Control module is well-architected for commercial applications, but faces significant environmental and connectivity risks in subterranean mechanical rooms. Hardware fail-safes and robust environmental sealing are required to ensure reliability.

Identified Issues

RISK
Firmware bugs could command unsafe setpoints yielding boiling or freezing conditions in the hydronic loop.

Suggestion: Incorporate hardwired, independent physical high-limit and low-limit mechanical thermostats that cut power or override software controls.

ENGINEERING
Wi-Fi and internal LTE antennas frequently fail in concrete basement plant rooms.

Suggestion: Add an SMA connector for an external mounted antenna and an RJ45 port for hardwired Ethernet fallback.

MATERIALS
Standard air-gap LCD screens are highly susceptible to delamination and condensation in humid mechanical environments.

Suggestion: Specify an optical bonding process for the LCD capacitive touch screen assembly and ensure an IP65 minimum rating for the bezel.

ENGINEERING
Cascading up to four 30kW units via Modbus risks signal collisions and ground loops over long cable runs.

Suggestion: Implement galvanically isolated RS-485 transceivers and include physical DIP switches for reliable line termination.

Data Sources:
[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

MANUFACTURABILITY

Humid environments will degrade exposed PCB components over the operational lifespan.

Suggestion: Apply conformal coating (acrylic or silicone) to the main microcontroller processing baseboard during PCBA.

Strategic Recommendations

- Integrate an SMA connector for an external LTE antenna and a standard RJ45 port for wired Ethernet.
- Specify an optical bonding process for the capacitive touchscreen and target [illegible] IP65+ enclosure to prevent moisture ingress.
- Incorporate independent mechanical high/low temperature limit switches in the unit to override dangerous firmware setpoints.
- Require conformal coating on the PCBA for reliability in humid plant rooms.
- Design the Modbus communication bus with isolated RS-485 transceivers and selectable termination resistors.

Data Sources:

[LLM] Gemini 3.1 Pro — Fang engineering review
[LLM] Gemini 3.1 Pro — cross-module consistency check

Modules — Quality Score

Section Evaluation: Modules

Composite Score: 8/10

Factual Integrity — are numbers accurate and claims grounded in reality?	8/10	Deterministic assessment
Visual Clarity — is the layout clean, readable, with proper hierarchy?	8/10	Deterministic assessment
Linguistic Precision — is the language professional, specific, free of filler?	8/10	Deterministic assessment
Logical Coherence — does this section flow from the previous one?	8/10	Deterministic assessment
Actionability — can the founder take concrete next steps from this section?	8/10	Deterministic assessment
Component Specification Clarity	8/10	Deterministic assessment
Interface Definition	8/10	Deterministic assessment
Failure Mode Depth	8/10	Deterministic assessment

5.1 BOM — Refrigerant Compression and Metering Assembly

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
outdoor_refrigeration_circuit-001	R290-Optimized Scroll Compressor	purchased_cots / varies	Purchased	-	£800.00
outdoor_refrigeration_circuit-002	Electronic Expansion Valve (EEV)	purchased_cots / varies	Purchased	-	£150.00
outdoor_refrigeration_circuit-003	4-Way Reversing Valve	purchased_cots / varies	Purchased	-	£80.00
outdoor_refrigeration_circuit-004	Suction Line Accumulator	purchased_cots / varies	Purchased	-	£40.00
outdoor_refrigeration_circuit-005	High/Low Pressure Transducers	purchased_cots / varies	Purchased	-	£60.00

5.2 BOM — Air-to-Refrigerant Evaporator Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
outdoor_evaporator_coil-001	Copper Micro-Groove Tubing	purchased_cots / varies	Purchased	-	£40.00
outdoor_evaporator_coil-002	Corrugated Aluminum Fins	sheet_metal / varies	Manufactured	-	£60.00
outdoor_evaporator_coil-003	Hydrophilic Anti-Corrosion Coating	purchased_cots / varies	Purchased	-	£40.00
outdoor_evaporator_coil-004	Defrost Base Heater Cable	purchased_cots / varies	Purchased	-	£40.00

5.3 BOM — Axial Fan & Shroud Module

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
outdoor_fan_assembly-001	Electronically Commutated (EC) Axial Fans	purchased_cots / varies	Purchased	-	£250.00
outdoor_fan_assembly-002	Aerodynamically Optimized Cowling/Shroud	sheet_metal / varies	Manufactured	-	£60.00
outdoor_fan_assembly-003	Vibration Isolation Motor Mounts	purchased_cots / varies	Purchased	-	£40.00
outdoor_fan_assembly-004	Bird Guard/Safety Grille	sheet_metal / varies	Manufactured	-	£60.00

5.4 BOM — Brazed Plate Heat Exchanger (Condenser)

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
outdoor_heat_exchanger-001	316L Stainless Steel Brazed Plate Heat Exchanger (BPHE)	purchased_cots / varies	Purchased	-	£600.00
outdoor_heat_exchanger-002	Polyurethane Thermal Insulation Jacket	purchased_cots / varies	Purchased	-	£15.00
outdoor_heat_exchanger-003	Water Flow Switch	purchased_cots / varies	Purchased	-	£40.00
outdoor_heat_exchanger-004	Water Supply/Return Thermistors	purchased_cots / varies	Purchased	-	£40.00

5.5 BOM — Power Electronics & Inverter Drive Enclosure

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
outdoor_power_electronics-001	3-Phase Inverter Drive PCB	purchased_cots / varies	Purchased	-	£400.00
outdoor_power_electronics-002	Main Logic Control Controller	purchased_cots / varies	Purchased	-	£180.00
outdoor_power_electronics-003	ATEX-Rated Hermetically Sealed Enclosure	sheet_metal / varies	Manufactured	-	£120.00
outdoor_power_electronics-004	EMI/RFI Electrical Filters	purchased_cots / varies	Purchased	-	£40.00
outdoor_power_electronics-005	High-Voltage Contactor Relays	purchased_cots / varies	Purchased	-	£40.00

5.6 BOM — Structural Frame and Acoustic Jacketing

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
outdoor_chassis_acoustics-001	Z275 Galvanized Steel Structural Frame	sheet_metal / varies	Manufactured	-	£60.00
outdoor_chassis_acoustics-002	Polyester Powder-Coated Exterior Panels	sheet_metal / varies	Manufactured	-	£60.00
outdoor_chassis_acoustics-003	Mass-Loaded Vinyl Acoustic Compressor Wraps	purchased_cots / varies	Purchased	-	£800.00
outdoor_chassis_acoustics-004	Condensate Drainage Basepan	sheet_metal / varies	Manufactured	-	£60.00
outdoor_chassis_acoustics-005	Rubber Isolation Feet	purchased_cots / varies	Purchased	-	£40.00

5.7 BOM — R290 Leak Monitoring & Safety Purge System

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
r290_safety_system-001	Infrared Combustible Gas Sensors (NDIR)	purchased_cots / varies	Purchased	-	£60.00
r290_safety_system-002	Intrinsically Safe Relay Board	purchased_cots / varies	Purchased	-	£40.00
r290_safety_system-003	Emergency Purge Exhaust Fan	purchased_cots / varies	Purchased	-	£250.00
r290_safety_system-004	Solenoid Safety Lock-Out Valves	purchased_cots / varies	Purchased	-	£80.00

5.8 BOM — Indoor Hydronic Pump Station (Hydrobox)

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
indoor_hydrobox-001	Variable Speed Circulating Pump	purchased_cots / varies	Purchased	-	£250.00
indoor_hydrobox-002	Diaphragm Expansion Vessel	purchased_cots / varies	Purchased	-	£40.00
indoor_hydrobox-003	Pressure Relief Valve (PRV, 3-bar)	purchased_cots / varies	Purchased	-	£80.00
indoor_hydrobox-004	Automatic Air Vent (AAV)	purchased_cots / varies	Purchased	-	£40.00
indoor_hydrobox-005	Magnetic System Filter	purchased_cots / varies	Purchased	-	£40.00

5.9 BOM — Human-Machine Interface & Building Management Control

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
indoor_ui_controller-001	LCD Capacitive Touch Screen Interface	purchased_cots / varies	Purchased	-	£40.00
indoor_ui_controller-002	Microcontroller Processing Baseboard	purchased_cots / varies	Purchased	-	£180.00
indoor_ui_controller-003	Modbus/BACnet Transceiver Chips	purchased_cots / varies	Purchased	-	£40.00
indoor_ui_controller-004	Wi-Fi/Cellular Connectivity Module	purchased_cots / varies	Purchased	-	£40.00
indoor_ui_controller-005	Weather Compensation Thermistor Inputs	purchased_cots / varies	Purchased	-	£40.00

BOM — Quality Score

Section Evaluation: BOM

Composite Score: 4/10

Criterion	Score	Comment
Factual Integrity — are numbers accurate and claims grounded in reality?	3/10	Every single component has a listed mass of 0kg, which represents a complete breakdown of physical realities for a hardware BOM.
Visual Clarity — is the layout clean, readable, with proper hierarchy?	5/10	The section is presented as a raw data dump of comma-separated key-value pairs rather than a structured table, making it difficult to read and process.
Linguistic Precision — is the language professional, specific, free of filler?	7/10	The descriptive names of the parts are technical and highly pertinent (e.g., 'ATEX-Rated Hermetically Sealed Enclosure'), although they serve as category descriptions rather than precise part specifications.
Logical Coherence — does this section flow from the previous one?	7/10	Components are logically organized and tagged by their parent modules (e.g., outdoor_refrigeration_circuit, r290_safety_system), showing a strong understanding of system architecture.
Actionability — can the founder take concrete next steps from this section?	3/10	Without part numbers, manufacturers, specific dimensions, quantities, or weights, an engineering or procurement team cannot actually order or source anything from this list.
Cost Realism	4/10	While a few major component costs are grounded (e.g., $800 for an R290 scroll compressor), there is excessive repetition of generic numbers ($40/$60) and some wild inaccuracies (e.g., $800 for an acoustic compressor wrap).
Part Number Validity	3/10	There are zero supplier or manufacturer part numbers provided; the system only generated generic programmatic IDs.
Quantity Break Accuracy	1/10	No unit quantities, volume tiers, or quantity pricing brackets are included in this output whatsoever.

Recommended Changes for Next Round:

- Modify the BOM generation prompt to strictly output a Markdown table with standard engineering columns (Module, Component, Part Number, Qty, Cost/Unit, Mass).
- Fix the mass calculation node in the pipeline so it utilizes a material density and bounding-box heuristic (or specific lookup) to prevent 'mass=0kg' fallbacks.
- Inject a COTS retrieval step or RAG database lookup to supply real-world manufacturer part numbers (e.g., matching a Danfoss or Copeland part to the R290 compressor) instead of generic strings.
- Refine the cost estimation logic to incorporate realistic volume discounts and eliminate default placeholder costs (the recurring $40/$60 values).
- Update the pipeline to include mass values for each part, potentially by integrating with a material database or calculating from dimensions and density.

6. Cost Waterfall & Economics

UNIT COST
£8,152.50

TARGET CEILING
None

HEADROOM
N/A

Make vs Buy Split

PURCHASED (BUY) TOTAL
£4,955.00

CUSTOM (MAKE) TOTAL
£480.00

MAKE %
5.9%

Per-Module Roll-up

MODULE	% OF UNIT	COST
Module outdoor_refrigeration_circuit	20.8%	£1,695.00
Module outdoor_evaporator_coil	3.3%	£270.00
Module outdoor_fan_assembly	7.5%	£615.00
Module outdoor_heat_exchanger	12.8%	£1,042.50
Module outdoor_power_electronics	14.4%	£1,170.00
Module outdoor_chassis_acoustics	18.8%	£1,530.00
Module r290_safety_system	7.9%	£645.00
Module indoor_hydrobox	8.3%	£675.00
Module indoor_ui_controller	6.3%	£510.00

Non-recurring engineering and overheads

Total non-recurring engineering (Tooling, Certs)	**£5,400.00**

Data Sources:
[Deterministic] Domain overhead multiplier model

Complexity Overhead Multiplier	**1.5×**

Grand Total (Unit plus Non-recurring Engineering) **£13,552.50**

Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

Data Sources:
[Deterministic] Domain overhead multiplier model

Cost — Quality Score

Section Evaluation: Cost

Composite Score: 4/10

Factual Integrity — are numbers accurate and claims grounded in reality?	6/10	System module costs are plausible for an R290 heat pump, but completely lack units (e.g., USD, EUR) and sourcing basis.
Visual Clarity — is the layout clean, readable, with proper hierarchy?	3/10	Data is presented as a raw, comma-separated text dump with no tables, lists, or visual hierarchy whatsoever.
Linguistic Precision — is the language professional, specific, free of filler?	6/10	Phrasing is robotic and fragmented ('Per-module: Module...'). It lacks standard professional financial or engineering language.
Logical Coherence — does this section flow from the previous one?	5/10	Possesses zero narrative flow. It abruptly starts with 'Unit total' without defining what product, phase, or volume is being costed.
Actionability — can the founder take concrete next steps from this section?	3/10	A founder cannot make supply chain decisions without knowing the production volume, tooling breakdown, or identifying the main cost drivers.
Modeling Assumptions Transparency	3/10	Only an 'Overhead multiplier: 1.5x' is stated. Missing labor rates, manufacturing location, BOM vs Assembly breakdown, and yield assumptions.
Volume Scaling Accuracy	2/10	No volume tiers (e.g., prototype, 1k units, 10k units) are mentioned or modeled.
Sensitivity Analysis	1/10	Completely absent. No minimum/maximum bounds, error margins, or worst-case scenario modeling provided.

Recommended Changes for Next Round:

- Modify the report generation template to map 'Per-module' dictionary/array items into a structured Markdown table including columns for 'Module', 'Cost', and '% of Total Cost'.
- Update the LLM prompt or data schema to strictly enforce the inclusion of 'currency', 'assumed_production_volume', and 'labor_location'.
- Add a post-processing pipeline step that automatically identifies the two highest-cost modules and generates a natural-language sub-section on cost reduction strategies.
- Introduce a sensitivity analysis function in the modeling pipeline that calculates +/- 15-20% variances on critical components (e.g., power electronics, heat exchangers) and outputs a risk breakdown.
- Modify the report generation pipeline to use structured tables for cost data, enhancing visual clarity and hierarchy, for example, by formatting module costs as a tabular list.

7.1 Refrigerant Compression and Metering Assembly Risks

Risk 1: Liquid slugging destroying compressor during low ambient temps

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Oversize the suction accumulator and tune EEV PID loops for conservative superheat.

Owner: Thermodynamics Engineer

Risk 2: Micro-leak of highly flammable R290 gas

Base Score: S:5 × L:3 = 15

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Automate brazing procedures in factory and implement 100% helium mass-spectrometer leak testing.

Owner: Manufacturing Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.1 Refrigerant Compression and Metering Assembly Risks

Risk 3: Compressor terminal ignition event

Base Score: S:5 × L:1 = 5

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Implement ATEX-approved sealed terminal covers and active base-pan ventilation.

Owner: Electrical Safety Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.2 Air-to-Refrigerant Evaporator Module Risks

Risk 4: Ice build-up destroying fin integrity

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Optimize hydrophilic coating and employ predictive AI defrost algorithms instead of time-based defrost.

Owner: Thermal Systems Engineer

Risk 5: Corrosion in coastal UK commercial deployments

Base Score: S:3 × L:4 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Apply a secondary epoxy e-coating (like Heresite) to the coil for aggressive environments.

Owner: Materials Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.2 Air-to-Refrigerant Evaporator Module Risks

Risk 6: Uneven airflow causing localized frosting

Base Score: S:3 × L:3 = 9

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Perform extensive CFD on fan placement vs coil geometry to guarantee uniform velocity profiles.

Owner: Aerodynamics Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.3 Axial Fan & Shroud Module Risks

Risk 7: Excessive tonal noise failing ErP and local planning regulations

Base Score: S:5 × L:3 = 15

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Partner with Tier 1 suppliers like ebm-papst for acoustically matched blade-and-shroud sets.

Owner: Acoustics/NVH Engineer

Risk 8: EC motor water ingress

Base Score: S:4 × L:2 = 8

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Specify strictly IP55 or higher rated EC motor enclosures with conformal coated electronics.

Owner: Mechanical Engineering Lead

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.3 Axial Fan & Shroud Module Risks

Risk 9: Ice-induced fan imbalance tearing the shroud

Base Score: S:4 × L:2 = 8

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Integrate RPM feedback algorithms to halt the fan if unexpected inertia/vibration spikes are detected.

Owner: Firmware Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.4 Brazed Plate Heat Exchanger (Condenser) Risks

Risk 10: BPHE rupture due to freezing

Base Score: S:5 × L:3 = 15

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Enforce a strict anti-freeze routine using the circulation pump, and mandate inline strainers/flow switches.

Owner: Hydraulics Engineer

Risk 11: Cross-contamination of R290 into building water supply

Base Score: S:5 × L:1 = 5

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Utilize highly engineered Tier-1 SWEP or Alfa Laval BPHEs with double-wall protection if required by local code.

Owner: Compliance Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.4 Brazed Plate Heat Exchanger (Condenser) Risks

Risk 12: Scale buildup dropping thermal efficiency

Base Score: S:3 × L:4 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Specify installation requirement for water treatment (magnetic filters, inhibitor fluids) in installation manuals.

Owner: Product Manager

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.5 Power Electronics & Inverter Drive Enclosure Risks

Risk 13: Ignition of leaked R290 gas via relay arc

Base Score: S:5 × L:1 = 5

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Design an intrinsically safe, multi-gasketed IP65 secondary enclosure for all spark-generating components.

Owner: Electrical Safety Engineer

Risk 14: IGBT Overheating

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Bond the inverter heatsink to the cold suction line of the refrigeration circuit for active liquid cooling.

Owner: Electrical Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.5 Power Electronics & Inverter Drive Enclosure Risks

Risk 15: EMC Test Failure

Base Score: S:3 × L:3 = 9

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Conduct pre-compliance EMC emissions testing at alpha stage to identify filter adjustments early.

Owner: Compliance Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.6 Structural Frame and Acoustic Jacketing Risks

Risk 16: Basepan ice compilation destroying internal components

Base Score: S:5 × L:3 = 15

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Design an aggressively stamped dual-slope basepan with redundant large-diameter drain ports and trace heaters.

Owner: Mechanical Engineering Lead

Risk 17: Panel rattle failing acoustic noise standards

Base Score: S:3 × L:4 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Use automotive-style structural adhesives and captive anti-vibration latches instead of traditional sheet metal screws.

Owner: Acoustics/NVH Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.6 Structural Frame and Acoustic Jacketing Risks

Risk 18: Unit tipping or failing in high winds

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Ensure center of gravity is kept exceedingly low; provide engineered wind-load bracket mounting accessories.

Owner: Structural Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.7 R290 Leak Monitoring & Safety Purge System Risks

Risk 19: Catastrophic gas explosion

Base Score: S:5 × L:1 = 5

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Maintain strictly hardwired, dual-redundant safety loops that bypass software logic entirely.

Owner: Electrical Safety Engineer

Risk 20: False positive gas detection stranding heating in winter

Base Score: S:3 × L:4 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Use high-quality optical NDIR sensors over cheap catalytic bead sensors to prevent cross-interference drift.

Owner: Electronic Design Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.7 R290 Leak Monitoring & Safety Purge System Risks

Risk 21: Propane pooling in electrical contactor areas

Base Score: S:5 × L:2 = 10

CAUSE	CONSEQUENCE
Unknown	Unknown

EXISTING CONTROLS	PLANNED MITIGATION
None	Design physical cabinet baffling to guarantee gravity drainage of heavy R290 gas away from electronics.

Owner: Mechanical Engineering Lead

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.8 Indoor Hydronic Pump Station (Hydrobox) Risks

Risk 22: Insufficient flow rate causing high delta-T and outdoor unit lockouts

Base Score: S:4 × L:3 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Specify a high head-pressure pump and integrate an automatic bypass valve into the hydrobox.

Owner: Hydraulics Engineer

Risk 23: Water leakage inside utility room causing property damage

Base Score: S:3 × L:2 = 6

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Implement internal drip trays and a water presence sensor that shuts isolation valves.

Owner: Mechanical Engineering Lead

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.8 Indoor Hydronic Pump Station (Hydrobox) Risks

Risk 24: Acoustic resonance into office spaces through walls

Base Score: S:2 × L:4 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Mount the internal pump via heavily dampened rubber isolation grommets.

Owner: Acoustics/NVH Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

7.9 Human-Machine Interface & Building Management Control Risks

Risk 25: MCS Compliance failure due to incorrect weather compensation implementation

Base Score: S:5 × L:2 = 10

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Engage MCS testing labs via software-in-the-loop early in development to validate logic trees.

Owner: Firmware Engineer

Risk 26: Loss of connectivity blocking remote fleet monitoring

Base Score: S:3 × L:4 = 12

CAUSE
Unknown

CONSEQUENCE
Unknown

EXISTING CONTROLS
None

PLANNED MITIGATION
Install an external high-gain antenna port allowing routing of signal out of concrete basements.

Owner: Hardware Engineer

Data Sources:
[LLM] Gemini 3.1 Pro — FMEA analysis

7.9 Human-Machine Interface & Building Management Control Risks

Risk 27: Cascade logic entering a hunting/oscillation state

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Develop robust master/slave token-ring PID logic with generous deadbands for stage increments.

Owner: Control Systems Engineer

Data Sources:

[LLM] Gemini 3.1 Pro — FMEA analysis

Risks — Quality Score

Section Evaluation: Risks

Composite Score: 8/10

Factual Integrity — are numbers accurate and claims grounded in reality?	8/10	Deterministic assessment
Visual Clarity — is the layout clean, readable, with proper hierarchy?	8/10	Deterministic assessment
Linguistic Precision — is the language professional, specific, free of filler?	8/10	Deterministic assessment
Logical Coherence — does this section flow from the previous one?	8/10	Deterministic assessment
Actionability — can the founder take concrete next steps from this section?	8/10	Deterministic assessment
Risk Coverage Completeness	8/10	Deterministic assessment
Probability Calibration	8/10	Deterministic assessment
Mitigation Specificity	8/10	Deterministic assessment

Suppliers — Quality Score

Section Evaluation: Suppliers

Composite Score: 6/10

Factual Integrity — are numbers accurate and claims grounded in reality?	6/10	Deterministic assessment
Visual Clarity — is the layout clean, readable, with proper hierarchy?	6/10	Deterministic assessment
Linguistic Precision — is the language professional, specific, free of filler?	6/10	Deterministic assessment
Logical Coherence — does this section flow from the previous one?	6/10	Deterministic assessment
Actionability — can the founder take concrete next steps from this section?	6/10	Deterministic assessment
Supplier Qualification Realism	6/10	Deterministic assessment
Geographic Viability	6/10	Deterministic assessment
Redundancy Evaluation	6/10	Deterministic assessment

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	**a_compact_commercial_airsource_heat_pump**
Research Result	**Generated**
System Modules	**9 Modules**
Bill of Materials Lines	**41 Parts**
Cost Analysis	**Complete**
Specialist Reviews	**9 Completed**
Supplier Shortlists	**41 Matched**

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction
Modules	LLM	Unknown LLM	Gemini 3.1 Pro — module decomposition
BOM	DETERMINISTIC	deterministic	Deterministic expansion from Max keyParts
BOM	LLM	Unknown LLM	Gemini 3.1 Pro — gap-fill for missing standard hardware
Cost	DETERMINISTIC	deterministic	Domain overhead multiplier model
Suppliers	SEARCH	search	Brave Search API — commercial supplier matching
Reviews	LLM	Unknown LLM	Gemini 3.1 Pro — Fang engineering review
Proofreader	LLM	Unknown LLM	Gemini 3.1 Pro — cross-module consistency check